NÉSTOR GARCÍA

Mechanical Construction Manager | Data Centre & Mission-Critical Mechanical Systems

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Available for new projects | Open to international rotation | Languages: Spanish (C2), English (C1), French (B1), Portuguese (B1), Italian (A2)

PROFESSIONAL SUMMARY

Mechanical Construction Manager with 23+ years and 34 projects delivering complete mechanical packages on large-scale industrial and critical infrastructure in 10 countries across Europe, Africa, the Middle East and North America. Experience spans the full package lifecycle: design coordination, procurement and material control, subcontractor management, installation, testing, punch lists, commissioning support and handover.

Direct hands-on delivery of the mechanical systems at the core of a hyperscale data centre: chilled-water and cooling plant (chillers, cooling towers, economizers) on 17 projects, fire protection on 17, water treatment on 12, and industrial HVAC, heat exchange, pumping and process piping. Responsible for final warranty sign-off on 19 projects and for 17 complex revamps in live operating plants with zero tolerance for downtime.

Worked for EPC contractors, specialist vendors and as Client Representative on design-and-build and turnkey projects with Técnicas Reunidas, Worley, Fluor, Foster Wheeler, Technip Energies, SENER and IDOM. Combines site leadership with strong commercial and data skills: cost and variation control, progress reporting, Power BI / SQL / Excel tracking of MTO, QA/QC and punch-list status, and AI-assisted document review and reporting.

CORE COMPETENCIES

- **Package management:** full mechanical package delivery, design coordination, design-and-build and turnkey projects, interface management with civil, electrical and commissioning teams
- **Commercial control:** schedule, resources and cost control (labour, subcontractors, materials, plant), variations and valuations, change management, weekly commercial reporting
- **Site delivery:** subcontractor management, material delivery scheduling and tracking, MTO control, progress reporting, site meetings and handovers
- **QHSE:** site safety leadership, QA/QC, ITPs, weld inspection, pressure testing, NCR management, snagging and punch lists
- **Close-out:** mechanical completion, O&M manuals, QA and safety files, documentation and photographic records, warranty sign-off
- **Commissioning support:** flushing, pressure testing, functional testing, start-ups, hydraulic balancing, ΔT validation
- **Mechanical systems:** chilled-water systems, chillers, cooling towers, economizers, CRAH/CRAC, heat exchangers and heat recovery, industrial HVAC and ductwork, fire protection, water treatment, pumping systems, compressed air, process piping, pipe racks and supports

DATA, DIGITAL TOOLS & AI

- **Project database:** structured 25 years of engineering documentation from 34 projects (about 140,000 files, 131 GB) into a relational SQL database of projects, systems, equipment, vendors, standards and key documents, connected to Power BI.
- **Online portfolio:** nestorgarcia-mcm.github.io, an interactive record of projects, systems and results built from that database.
- **Project controls analytics:** Power BI models (DAX, Power Query) for MTO vs progress, weld book reconciliation, QA/QC and punch-list status.
- **AI-assisted workflows:** Claude, ChatGPT and Microsoft 365 Copilot for specification, datasheet and contract review, correspondence triage, reporting and data extraction, using reusable instructions and scheduled agents with human sign-off.

DATA CENTRE & MISSION-CRITICAL MECHANICAL EXPERIENCE

Transferable mechanical delivery experience on the systems that define a hyperscale data centre, built across power plants, refineries and process facilities where continuous operation is critical.

- **Cooling plant – chillers, cooling towers, economizers:** 17 projects, 17 years. Chilled-water and condenser-water networks, heat rejection, pumping and controls interfaces (Kathu CSP 100 MW, EDF Réunion 108 MW, Viridor Avonmouth ERF, REPETCO).
- **Heat exchange and heat recovery:** heat exchangers, thermal fluid and steam/condensate systems, waste-heat recovery and heat rejection on power and process plants – directly relevant to data centre waste-heat export.
- **Fire protection:** 17 projects, 18.5 years. Pump houses, hydrant and sprinkler networks, water storage, firefighting systems to plant safety regulations.
- **Water treatment:** 12 projects, 13.1 years. Make-up water, filtration, purification and recirculation systems (WTP / WWTP).
- **Mission-critical HVAC and ductwork:** 3 projects, 4.25 years. Technical rooms, industrial ventilation and air-quality systems (REPETCO, Abengoa Lacq and Salamanca).
- **Critical power interfaces:** UPS and battery systems on 5 projects (6.75 years); standby diesel generation on 4 projects; bulk fuel storage and transfer on 8 projects.
- **Redundancy and uptime:** installation and testing of N+1 / N+2 mechanical configurations; 17 brownfield revamps executed in live plants under strict permit-to-work and outage windows – equivalent to works in live data halls.
- **Commissioning:** pre-commissioning, flushing, pressure testing, functional testing, hydraulic balancing and ΔT validation, supporting Cx teams through to handover.

PROFESSIONAL EXPERIENCE

Senior Mechanical Construction Manager | Aldesa Ingeniería (Contractor) | 2025 – 2026

Moeve Picasso Project, La Rábida Energy Park, Palos de la Frontera, Spain | Engineering: Worley | Greenfield

- Managed the mechanical package for 2 units of Southern Europe's largest 2G biofuels complex (500,000 t/yr SAF and HVO).

- Directed prefabrication and installation of 250,000 pipe-welding inches, static and rotating equipment, structures and insulation, with modular pre-assembled packages.
- Controlled subcontractors, material deliveries, progress, QA/QC, punch lists and mechanical completion; systems included cooling plant, fire protection, pumping and compressed air.
- Built a Power BI model crossing 35,017 engineering and SAP material codes to forecast welding inches before material arrived, and reconciled the weld book against monthly certification.

Project Manager / Senior Mechanical Specialist | Sulzer Chemtech – Tower Field Services (Vendor) | 2023 – 2025

10 projects in Spain, Portugal, Italy and Hungary for Repsol, BP, Dupont DOW, ENI, BorsodChem, Galp and Rogerio Leal | Brownfield and greenfield

- Managed simultaneous revamp and turnaround packages for reactors and separation columns: installation, QA/QC, hydraulic tolerance checks and mechanical completion.
- Signed performance guarantees on all 10 projects and 5-year warranties on weld overlay works and repairs.
- Delivered projects weeks ahead of schedule, increasing client satisfaction by 50%, with full API and ASME compliance.

Mechanical Construction Manager | Técnicas Reunidas (Client Representative) | 2022 – 2023

REPETCO multilayer PET/PE recycling plant, Albacete, Spain | Greenfield

- Managed the complete mechanical package: process packages, pressure vessels and heat exchangers, industrial HVAC and ventilation, fire protection, water treatment, compressed air, UPS interfaces and storage silos.
- Resolved site issues to prevent delays and applied time, resource and cost controls, completing under budget; signed final warranty.

Mechanical Construction Supervisor | Repsol (Client Representative) | 2022

Petronor Bilbao refinery (desulfurization, hydrocracker, coker) and Tarragona naphtha unit revamps, Spain | Engineering: Técnicas Reunidas

- Supervised mechanical revamp works in live refineries, including cooling, fire protection, fuel and water treatment systems, through to final warranty sign-off.

Mechanical Construction Supervisor | BASF (Client Representative) | 2021 – 2022

Propane dehydrogenation (PDH) plant revamp, Tarragona, Spain | Licensor: UOP Honeywell

- Coordinated mechanical equipment installation and testing for reactors and separation columns with zero critical NCRs.

Mechanical Construction Supervisor | Technip Energies Benelux (Vendor) | 2021

F-3201 hydrogen unit full revamp, BP Castellón refinery, Spain

- Managed static and dynamic equipment installation with full API and ASME compliance; signed final warranty.

Site Manager | Axial Structural Solutions (Contractor) | 2020 – 2021

18 MW photovoltaic plant (27,000 panels) for EDF / ENGIE, Fanjeaux, France | Greenfield

- Led site execution, subcontractors and completion, including fire protection, standby power and UPS / battery interfaces.

Site Manager | KAEFER (Contractor) | 2019 – 2020

Avonmouth Energy Recovery Facility for Viridor, Bristol, United Kingdom | Engineering: CNIM | Greenfield

- Managed site delivery of the mechanical scope, including cooling plant and fire protection, with workshop prefabrication and QA/QC.

Mechanical Construction Supervisor | INEOS Inovyn (Client Representative) | 2018 – 2019

VCM and PVC production plant revamp, Martorell, Spain | Engineering: IDOM

- Supervised mechanical revamp works including cooling, pumping and process piping; signed final warranty.

QA/QC Manager | Maracof Industrial (Contractor) | 2017 – 2018

Kathu 100 MW Concentrated Solar Power plant with molten-salt storage, South Africa | Engineering: SENER | Greenfield

- Led QA/QC for the mechanical package: cooling plant, water treatment, fire protection, fuel, compressed air, pumping and UPS systems.

Mechanical Construction Manager | HAMON – Esindus (Vendor) | 2015 – 2016

40 MW biomass power plant for Veolia, Merritt, Canada | Engineering: Iberdrola Ingeniería | Greenfield

- Managed the vendor mechanical package including cooling plant, fire protection and pumping; signed final warranty.

Mechanical Construction Manager | Elecnor – Hidroambiente (Vendor) | 2014 – 2015

SKS Steel iron-making plant water purification and irrigation systems, Bandar Abbas, Iran | Engineering: IDOM | Greenfield

- Managed the water treatment and cooling package, including standby power and fire protection; signed final warranty.

Site Manager | UTE Enwesa Maesa (Contractor) | 2012 – 2014

108 MW diesel power plant (MAN 18V48/60 engines, 18.5 MW each) for EDF, PEI Port, Réunion Island, France | Greenfield

- Led site execution of the mechanical package: cooling, fuel oil storage and transfer, fire protection, water treatment and compressed air.

Contract Manager | Nervion Industries (Contractor) | 2011 – 2012

C10 Cartagena refinery expansion for Repsol, Spain | Engineering: Fluor | Greenfield

- Managed contracts, variations and site installation of mechanical works across new process and auxiliary units.

Contract Manager | Abengoa (Engineering) | 2009 – 2010

Bioethanol refinery, Lacq, France | Greenfield

- Managed mechanical contracts and installation, including cooling plant, HVAC, fire protection, water treatment and UPS systems.

Mechanical Construction Supervisor | Técnicas Reunidas (Engineering) | 2008 – 2009

Rabigh Development Project refinery units for Saudi Aramco, Saudi Arabia | Greenfield

- Supervised construction and erection of new process units with cooling, fuel, water treatment and fire protection systems.

Mechanical Construction Supervisor | Avantia TICSA (Contractor) | 2007 – 2008

900 MW combined cycle power plant La Plana del Vent, Tarragona, Spain | Engineering: Técnicas Reunidas | Greenfield

QA/QC Manager | Técnicas Reunidas (Client Representative) | 2005 – 2007

Bioethanol refinery for Ebro Puleva, Babilafuente, Salamanca, Spain | Engineering: Abengoa | Greenfield

- Led QA/QC for a package covering all mission-critical systems: cooling plant, HVAC, standby power, UPS, fuel, fire protection and water treatment; signed final warranty.

Contract Manager | KAEFER (Contractor) | 2004

Nylon 6 plant for UBE, Castellón, Spain | Engineering: Técnicas Reunidas | Greenfield

Technical Office, Warehouse Manager and QA/QC Manager | Sodes S.A. (Contractor) | 2001 – 2004

Repsol Tarragona hydrocracker; 400 MW combined cycle plant, Tarragona; BP Castellón refinery revamp; 250 MW Cercs and 400 MW Bahía de Algeciras coal power plant revamps, Spain | Engineering: Técnicas Reunidas, Foster Wheeler

KEY PROJECTS

Years	Project and client	Location	Scale	Role	Key mechanical systems
2025 – 2026	Picasso 2G biofuels complex – Moeve (Aldesa, Worley)	Spain	500,000 t/yr; 250,000 pipe-welding inches	Senior Mechanical Construction Manager	Process piping, cooling plant, fire protection, pumping, compressed air

Years	Project and client	Location	Scale	Role	Key mechanical systems
2023 – 2025	10 refinery and petrochemical revamps – Sulzer Chemtech	Spain, Portugal, Italy, Hungary	10 turnarounds	Project Manager	Reactors and columns, piping, structures; warranty sign-off on all 10
2022 – 2023	REPETCO PET/PE recycling plant – Técnicas Reunidas	Spain	Complete greenfield plant	Mechanical Construction Manager	Cooling plant, HVAC, heat exchangers, fire protection, water treatment, UPS, standby power
2019 – 2020	Avonmouth Energy Recovery Facility – Viridor (KAEFER, CNIM)	United Kingdom	Waste-to-energy power plant	Site Manager	Cooling plant, fire protection, structures
2017 – 2018	Kathu Concentrated Solar Power plant – ENGIE (Maracof, SENER)	South Africa	100 MW, molten-salt storage	QA/QC Manager	Cooling plant, water treatment, fire protection, fuel, UPS
2015 – 2016	Merritt biomass power plant – Veolia (HAMON)	Canada	40 MW	Mechanical Construction Manager	Cooling plant, fire protection, pumping
2012 – 2014	PEI Port diesel power plant – EDF (Enwesa Maesa, MAN)	Réunion Island, France	108 MW	Site Manager	Cooling, fuel storage and transfer, fire protection, water treatment
2005 – 2007	Babilafuente bioethanol refinery – Ebro Puleva (Técnicas Reunidas)	Spain	Complete greenfield plant	QA/QC Manager	Cooling plant, HVAC, standby power, UPS, fuel, fire protection, water treatment

CERTIFICATIONS & TECHNICAL CODES

Certifications

- Inspector of Welded Construction, Level II – 07/2016

- CFR – Certified First Responder – 03/2020
- Mass Transfer Technologies – 08/2023

Codes and standards

- **Building services and HVAC:** ASME B31.9 (building services piping), ASME B31.5 (refrigeration piping), ASHRAE, EN 378 (refrigerating systems), RITE (Spanish building thermal installations regulation)
- **Piping:** ASME B31.1, B31.3, B31.4, B31.8, B31.11, B31.12, B31Q; ASME B16.5, B16.9, B16.20, B16.34
- **Pressure equipment:** ASME BPVC Sections I, II, V, VIII and IX; UNE-EN / DIN EN 13480
- **Rotating equipment and tanks:** API 610, API 650, API 12J; welding API 1104
- **Materials and pipelines:** ISO 3183, ISO 15156, UNE-EN 10208, UNE-EN 10255, UNE-EN ISO 21809, DIN EN ISO 19008
- **Management systems:** ISO 9001 (quality), ISO 14001 (environmental)
- **Materials experience:** carbon steel (23 projects), alloy steels P22 / P91 (29), stainless steel (31), Inconel, high-pressure components (27)

EDUCATION, LANGUAGES & SOFTWARE

Education

- B.Sc. Computer Databases IT – Universitat Politècnica de València, Spain – 2001
- Graduate Certificate of Education – I.B. Ramon Llull, Valencia, Spain – 1997

Languages

- Spanish – native / bilingual (C2)
- English – advanced (C1), daily working language on international projects
- French – intermediate (B1); Portuguese – intermediate (B1); Italian – elementary (A2)

Software

- **Project controls:** Oracle Primavera P6 (programme and progress tracking); Microsoft Excel – Power Query, VBA (cost, variations and valuations)
- **Construction management and BIM:** Autodesk Construction Cloud / BIM 360; Autodesk AutoCAD, Navisworks and Revit (model review, clash mitigation, quantity and data extraction)
- **Data and reporting:** Microsoft Power BI (DAX, Power Query, semantic models), SQL and SQLite databases, Python data scripts (AI-assisted), Oracle, Microsoft Access
- **AI tools:** Claude (projects, skills, connectors, scheduled agents), ChatGPT, Microsoft 365 Copilot; Canva for visual reporting
- **Office:** Microsoft Office (Word, Excel, Outlook, PowerPoint)
- **Engineering:** SAP2000